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ANALOGUE RC FILTERS

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1. INTRODUCTION TO ANALOGUE FILTERS

Analogue filters are frequency-selective circuits designed to modify, reshape or reject a specified band of frequencies while attenuating all those unwanted signals outside of this band. That is, electronic filters “filter-out” unwanted signals.

Electronic filters can be designed to be either passive or active networks. *Passive filter networks* have no amplifying elements, so produce no signal gain. They commonly contain only passive components such as resistors, capacitors and/or inductors. *Active filter networks* commonly use operational amplifiers (op-amps) for voltage amplification, as well as resistors and capacitors within their design.

Filter circuits work in the frequency domain and as such are named according to the frequency range of input signals that they allow to pass, while at the same time block or reject the rest. The frequency point at which an electronic filter passes or blocks a signal is called its corner or cut-off frequency, (f_c).

Electronic filters can be Passive or Active designs using resistors, capacitors, inductors or op-amps

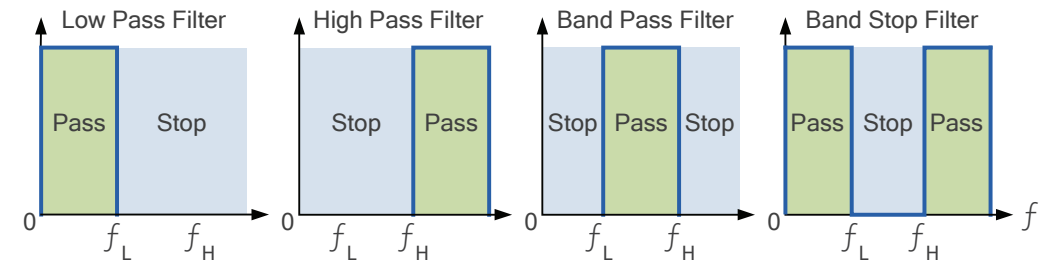
The four most commonly used filter designs (topologies) are the:

- **Low Pass Filter** – the low pass filter is the most common filter design. It allows only low frequency signals from 0Hz (DC) to its corner frequency point ($f \leq f_{c(LP)}$) to pass while blocking all the higher input signals.
- **High Pass Filter** – is the reverse of the low-pass filter by allowing only the high frequency signals to pass from its corner frequency point to infinity ($f \geq f_{c(HP)}$) while blocking those any lower.
- **Band Pass Filter** – this filter design allows input signals falling within a specified frequency band between two points to pass unaffected, while blocking both the lower and higher frequencies either side of this band. These upper and lower points are symmetrical around a centre frequency.
- **Band Stop Filter** - the band stop filter blocks or attenuates and frequency signals within a certain upper and lower band, while allowing all other signals to pass. The stop band can be narrow or broad depending on its selectivity.

As the function of any electronic filter is to allow signals of a given band or range of input frequencies to pass unaltered while attenuating or rejecting all those others which are not wanted. The range of frequencies which a particular filter topology allows to pass unaffected is called its **Pass Band**. The range of frequencies which are rejected is called the **Stop Band**.

We can define the frequency response characteristics of any analogue filter by using an ideal frequency response showing the four basic filter designs as shown in Figure 1.

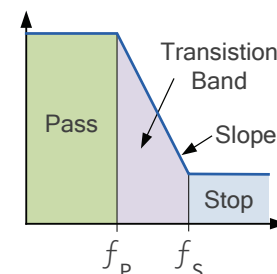
FIGURE 1. IDEAL FILTER RESPONSE



While the ideal filter response characteristics of Figure 1 shows an abrupt and vertical change from pass band to stop band, or the stop band back to the pass band depending upon the filter design.

For real analogue filters there is a region known as the filters **Transition Band** in which the change from pass band (f_p) to stop band (f_s), and vice versa is not vertical, but takes the form of a slope which is specified in decibels (dB).

FIGURE 2. THE TRANSITION BAND



The transition band shows the sharpness of the cut-off of the filter which, apart from other things, largely determines the complexity of the filter. Clearly the width of the transition between pass band and stop band should be as narrow as possible so as to prevent the passage of unwanted frequency signals through the particular filter.

Filters separate signals passing those of interest while rejecting the unwanted frequencies

Simple first-order (1st-order) passive filters can be made by connecting together a single resistor and a single capacitor in series across an input signal, (V_{IN}). The corresponding output signal, (V_{OUT}) is taken from the junction of these two components.

Depending on which way around we connect the resistor and the capacitor with regards to the output signal determines the type of filter construction producing either a simple Low Pass Filter or a simple High Pass Filter.

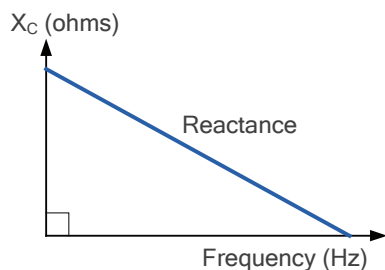
2. THE PASSIVE LOW PASS FILTER

A **Passive Low Pass Filter, (LPF)** circuit is nothing more than a series RC circuit in which the input signal is applied across the two series connected components, while the output signal is measured directly across the terminals of the capacitor.

Resistance, (R) and capacitance, (C) always have a constant value, for example $1k\Omega$ or $1\mu F$ respectively. However, capacitors also have another value called reactance, (X) which changes value with changes in applied frequency, f . Since we are dealing with capacitors, their reactance is called *Capacitive Reactance* (X_C) and is measured in ohms, (Ω).

As reactance is linked to frequency, it therefore follows that if the input signal frequency changes, X_C also changes causing the capacitors impedance to change. Then we can show the effect of frequency against capacitive reactance in Figure 3.

FIGURE 3. CAPACITIVE REACTANCE VS FREQUENCY



Note that even though the capacitors reactance changes with frequencies, its capacitance value in Farads will always remain the same.

Thus, the capacitive reactance of a capacitor at a particular frequency is given as:

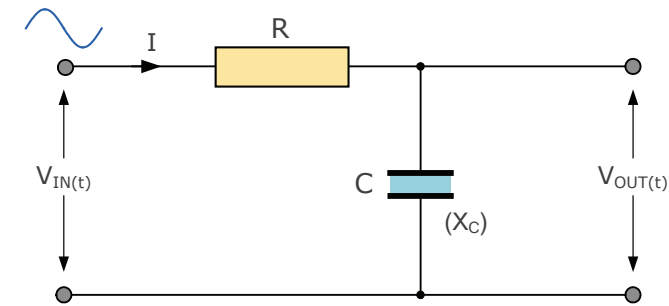
$$X_C = \frac{1}{2\pi f C}$$

Hence, the voltage drop, V_C across the capacitor equals: $X_C \cdot I_C$

If we now look at using the same series RC circuit in the frequency domain with a voltage source as V_{IN} (input voltage), and the voltage drop across the capacitor as V_{OUT} (output voltage). We can derive a transfer function to show the relationship between the input and output and therefore its operation as a passive low pass filter.

A low pass filter allows all low frequency signals to pass from DC (0Hz) up to its cut-off or corner frequency, (f_c) and blocks all high frequency signals above this point. A first-order RC passive low pass filter is constructed as shown in Figure 4.

FIGURE 4. PASSIVE LOW PASS RC FILTER CIRCUIT



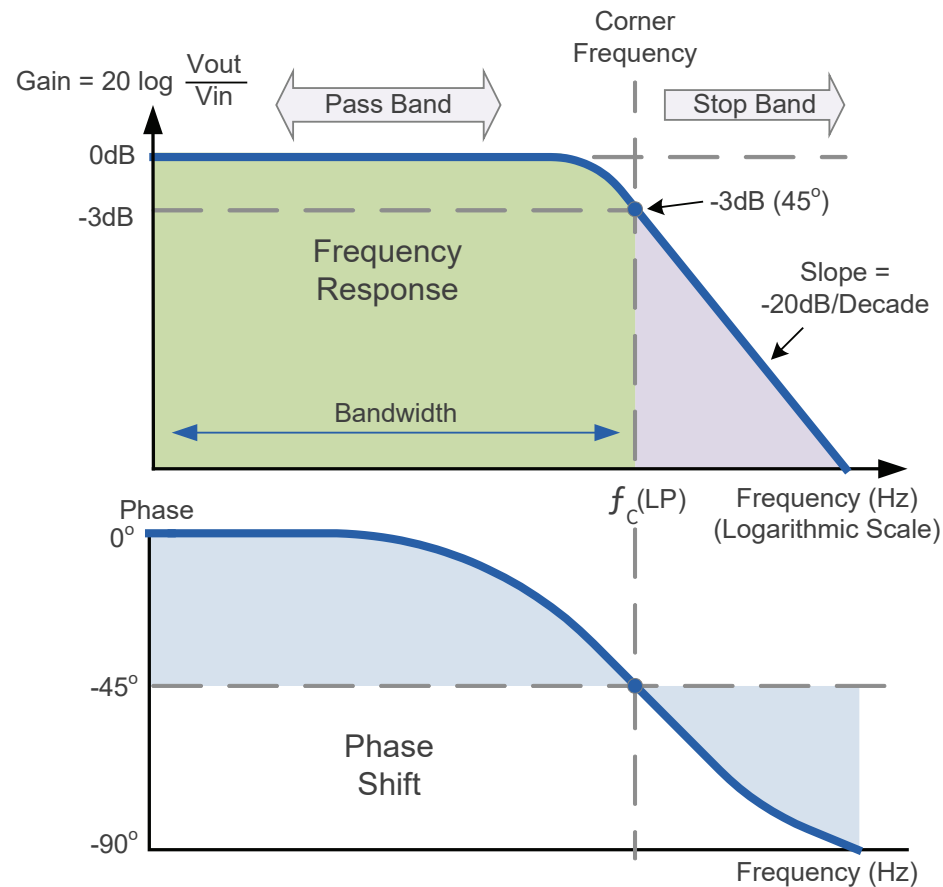
The reactance of the capacitor in Figure 4 varies inversely with frequency, while the value of the resistor remains constant. At low input frequencies the reactance of the capacitor is very large compared to the resistive value of the resistor. So the output voltage across the capacitor will be much larger than the I^2R voltage drop developed across the resistor.

At high frequencies the reverse is true with V_C being small and V_R being large due to the change in the capacitive reactance value. Then the passive RC low pass filter circuit acts as a frequency dependant variable potential divider circuit with the output voltage being controlled by the input signal frequency.

Low pass filters pass a group or band of frequencies between DC and its corner frequency

By plotting the networks output voltage against different values of input frequency, the frequency response curve or Bode plot function of the pass low pass filter circuit can be constructed as shown in Figure 5.

FIGURE 5. LOW PASS FILTER FREQUENCY RESPONSE CURVE



The *Bode Plot* of Figure 5 shows the frequency response of the filter to be nearly flat for low frequencies. Thus all of the input signal is passed directly to the output resulting in a gain of nearly 1, called unity. This continues until it reaches the corner or cut-off frequency point of the filter at its $f_{c(LP)}$ point.

After this corner frequency point the response of the circuit decreases to zero at a slope of -20dB/ Decade or (-6dB/Octave) “roll-off”. Note that the angle of the slope, this -20dB/decade roll-off, will always be the same for any 1st-order filter combination.

Any input signal with a frequency above this corner frequency point will become greatly attenuated due to the low reactance of the capacitor. This gives the effect of a short circuit condition on the output terminals resulting in zero output.

Then by carefully selecting the correct resistor-capacitor (RC) combination, we can create a series RC filter circuit that allows a chosen range of frequencies below a certain value to pass through unaffected while rejecting any input frequencies applied to the circuit above this cut-off point creating our low pass filter circuit.

The corner, or cut-off frequency point is defined as being the frequency where the ohmic value of the capacitive reactance and the resistance are equal. That is: $X_c = R$.

When this occurs the output signal is attenuated to **70.7%** of the original input signal value or **-3dB** ($20\log(V_{out}/V_{in})$) of the input. A negative decibel value means that the output voltage is less than the input voltage, *attenuation*. Although $X_c = R$, the output voltage is not half of the input as it would in a fully resistive voltage divider network since it is equal to the vector sum of the two voltages and therefore $V_{OUT} = 0.707V_{IN}$ of the input.

The filters corner frequency point can be found by using the following standard equation:

$$f_c = \frac{1}{2\pi RC} \text{ Hz}$$

As the passive low pass filter circuit contains a capacitor, the *Phase Angle* (Φ) of the output signal “lags” behind that of the input. Thus at the **-3dB** corner frequency (f_c) it is **-45°** out-of-phase. This is due to the output voltage (the voltage across the capacitor) “lagging” behind that of the input signal. Then the higher the input frequency the more the capacitor lags and the circuit becomes more and more “out of phase”.

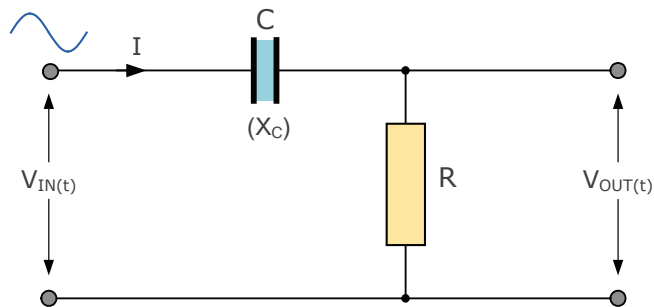
The filters phase shift angle is given as: **Phase Angle (ϕ) = $-\tan^{-1}(2\pi fRC)$**

Thus, a passive RC low pass filter circuit consisting of a **1.6kΩ** resistor and a **10nF** capacitor would have a phase shift angle of **-45°** at a corner frequency of **10kHz**.

3. THE PASSIVE HIGH PASS FILTER

Whereas the previous low pass filter allowed signals to pass below its corner frequency, -3dB point. The **Passive High Pass Filter, (HPF)** circuit as its name implies, passes all those input signals above the selected corner frequency while eliminating the ones below it. A simple first-order high pass RC filter circuit is shown in Figure 6.

FIGURE 6. PASSIVE HIGH PASS RC FILTER CIRCUIT



We can see here that the resistor and capacitor positions have been interchanged. As we saw previously, the reactance of a capacitor is very high at low frequencies. The capacitor acts like an open circuit and blocks any low frequency input signals, V_{IN} , thus the voltage across the capacitor, V_C is large while the voltage across the resistor, V_R is small.

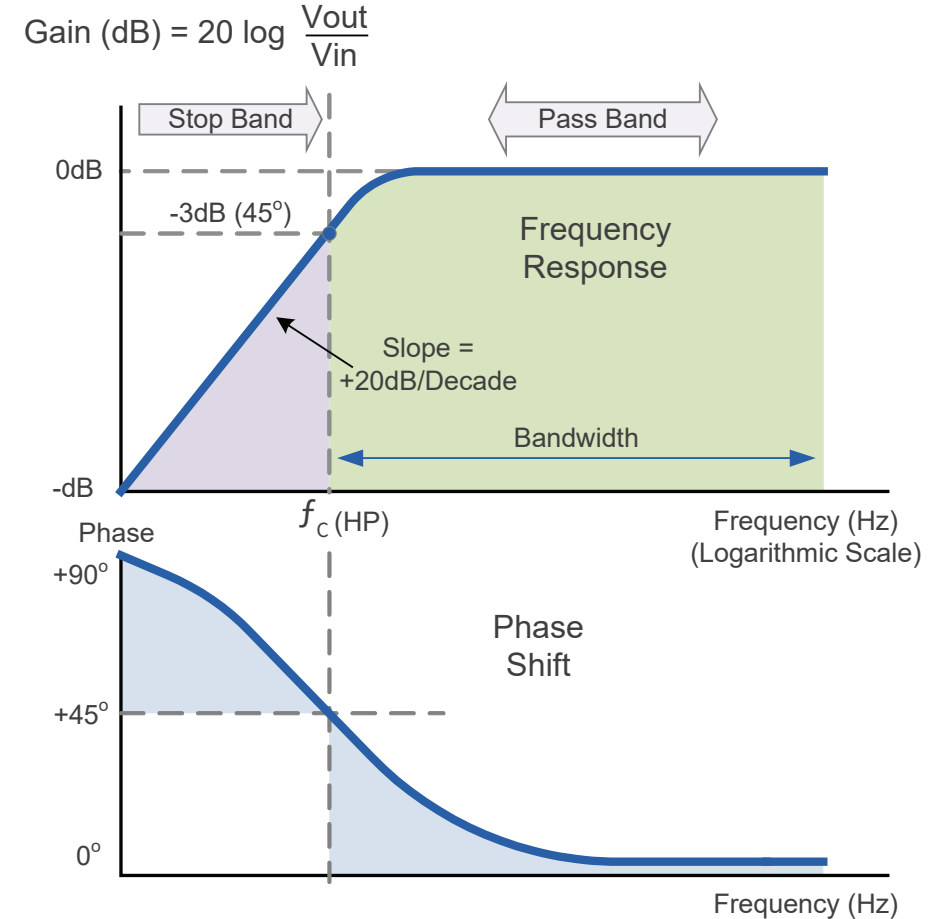
As the input frequency increases, the reactance of the capacitor decreases, so V_C reduces while V_R increases. That is the output voltage increases. So the higher the input frequency, the higher the output voltage producing the characteristics of a passive high pass filter.

Again as with the previous low pass filter circuit. The -3dB corner or cut-off frequency point is given by the following standard equation.

$$f_c = \frac{1}{2\pi RC} \text{ Hz}$$

Note that above this corner frequency point the reactance of the capacitor, X_C has reduced sufficiently as to now act more like a short circuit allowing all of the input signal to pass directly to the output as shown below in the filters response curve of Figure 7.

FIGURE 7. HIGH PASS FILTER FREQUENCY RESPONSE CURVE



The Bode Plot or Frequency Response Curve of Figure 7 is the exact opposite to that of a low pass filter. Here the input signal is attenuated or damped at low frequencies with the output increasing at +20dB/decade (6dB/Octave) until the frequency reaches the corner point ($f_{c(HP)}$) where again $R = X_C$.

For the high pass filter, the *Phase Angle* (Φ) between the input and the output signal “leads” that of the input and is equal to $+45^\circ$ at the -3dB point. The equation for the phase angle of the high pass filter circuit is modified slightly to that of the low pass filter circuit to account for the positive phase angle and is given as:

$$\text{Phase Angle } (\phi) = \tan^{-1} 1/(2\pi fRC)$$

Then for high frequencies the phase shift angle will approach 0° (zero degrees), since more of the input signal passes directly to the output the higher the frequency.

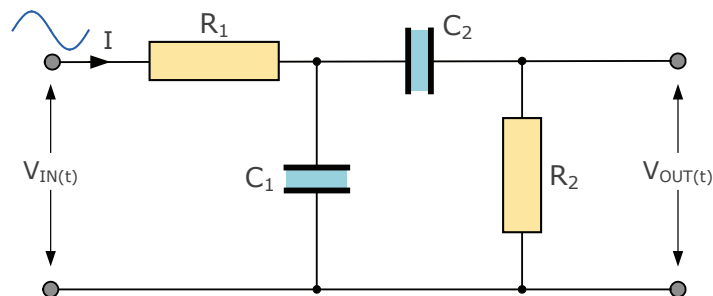
Note that the frequency response curve for the high pass filter implies that the filter can pass “all” signals to infinity (and beyond!). However, in practice the response of the filter can not extend up to infinity as it is limited by the electrical characteristics of the resistive and capacitive components used.

4. THE PASSIVE BAND PASS FILTER

If we take a single low pass filter and cascade it together with a single high pass filter, we can create another type of passive RC filter topology called a **Band Pass Filter, (BPF)** that allows the input signal to pass through a specified frequency range or band.

The band pass filter uses two resistors and two capacitors within its design whose values can be changed to tune the width of the desired pass band as shown in Figure 8.

FIGURE 8. PASSIVE BAND PASS RC FILTER CIRCUIT

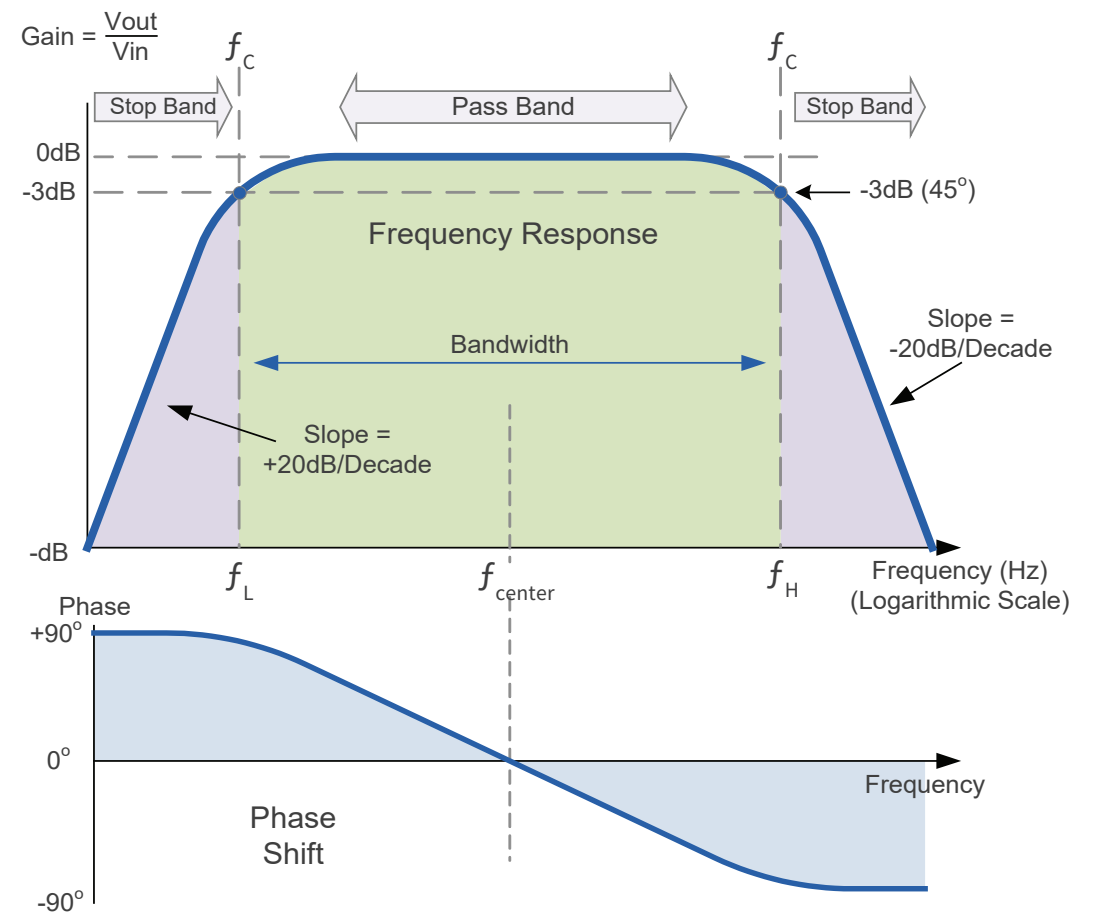


The low pass response is obtained by R_1 and C_1 , while the high pass response is R_2 and C_2 .

A band pass filter blocks all low frequencies below its lower corner frequency, (f_L) and all high frequency above its upper corner frequency, (f_H) only allowing frequency signals in between the band (range) to pass. Then a band pass filter has two corner frequencies.

Clearly, the corner frequency of the low pass filter must be greater than that of the high pass filter and the difference between the two frequencies at the -3 dB point determining the “bandwidth” of the band pass filter as shown in Figure 9.

FIGURE 9. BAND PASS FILTER FREQUENCY RESPONSE CURVE



The Bode Plot or frequency response curve of Figure 9. shows the output characteristics of the band pass filter. Here the signal is attenuated at low frequencies with the output increasing at a slope of +20dB/decade until the frequency reaches the “lower cut-off” point f_L . At this frequency the output voltage is again $1/\sqrt{2} = 70.7\%$ of the input signal value or -3dB ($20 \cdot \log(V_{OUT}/V_{IN})$) of the input.

The output continues at maximum gain until it reaches the “upper cut-off” point f_H where the output decreases at a rate of -20dB/decade attenuating any high frequency signals. The point of maximum output gain is generally the geometric mean of the two -3dB value between the two corner points.

A band pass filter is regarded as a second-order (two-pole) type filter because it has “two” reactive components (capacitors) within its circuit design. Thus, the phase shift angle will be twice that of the previously seen first-order filters, ie, 180°.

Band pass filters pass a group or band of frequencies between two corner frequency points

The phase angle of the output signal LEADS that of the input by +90° up to the centre point where it becomes “zero” degrees (0°) or “in-phase” and then changes to LAG the input by -90° as the output frequency increases.

The upper and lower cut-off frequency points for a band pass filter can be found using the same formula as that for both the previous low and high pass filters. That is:

$$f_c = \frac{1}{2\pi RC} \text{ Hz}$$

Then clearly, the width of the pass band of the filter can be controlled by the positioning of the two cut-off frequency points of the two RC filters.

We can calculate the center frequency (f_{center}) point of the band pass filter were the output is at its peak value. This peak value is not the arithmetic average of the upper and lower -3dB corner points as you might expect, but is in fact the geometric value of the two.

$$f_{\text{center}} = \sqrt{f_L \times f_H}$$

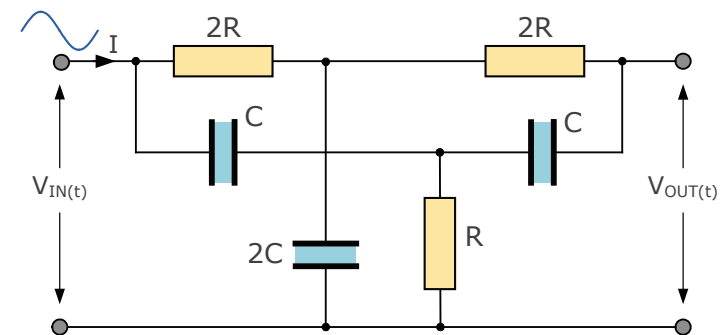
Where, f_{center} is the center frequency, f_L is the lower -3dB corner frequency point, and f_H is the upper -3db corner frequency point.

5. THE PASSIVE BAND STOP FILTER

The **Band Stop Filter, (BSF)** also known as a *Notch Filter* or *Bandreject Filter*, is another type of filter design which has two corner or cut-off frequencies. The band stop filter blocks (rejects) or at least severely attenuates, a certain band of frequencies inbetween its two corner frequencies while allowing those outside of this stop band to pass unaffected.

Thus the frequency response of the band pass filter is the exact opposite of the previous band pass filter of Figure 9. and again is constructed by cascading together a low and a high pass filter circuit to form what is called a twin-T configuration as shown in Figure 10.

FIGURE 10. PASSIVE BAND STOP RC FILTER CIRCUIT



Then components 2R and 2C form the low pass filter section, and R and C form the high pass filter section. Note that 2R means twice the value of R, and 2C twice the value of C.

Passive RC band stop, or notch filters where the gain is unity everywhere either side of the notch can be useful. A passive twin-T band stop filter could be used to attenuate or reject one or more undesirable frequencies.

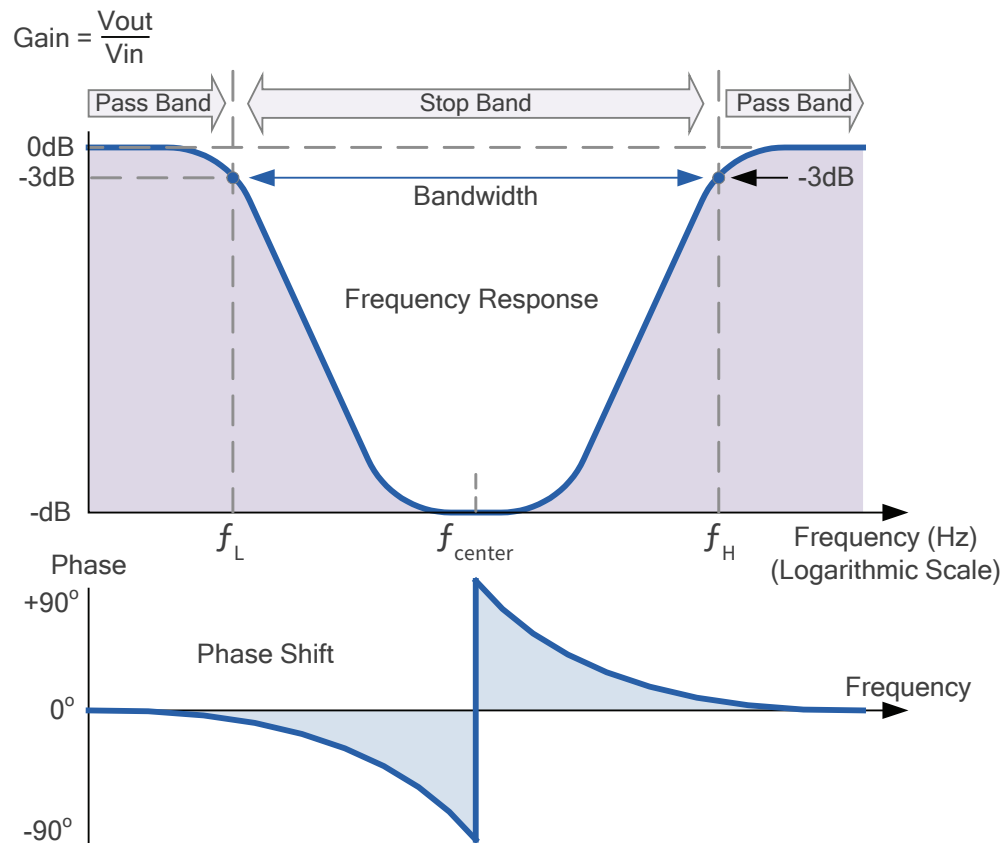
Band stop filters can be used to block or reject a single or band of unwanted signals between two corner frequency points

For example, it may be necessary to attenuate or reject a specific frequency generating electrical noise, such as 50Hz or 60Hz mains hum or the removal of unwanted harmonics, etc.

Then the band stop filter can be used to attenuate a single or very small narrow band of frequencies rather than a wide bandwidth of different frequencies.

For a wide-band band stop filter, its actual stop band lies between its lower and upper -3dB points attenuating any unwanted signals between these two corner frequencies. The frequency response curve of an ideal band stop filter is given in Figure 11.

FIGURE 11. BAND STOP FILTER FREQUENCY RESPONSE CURVE



We can see from the amplitude and phase curves of Figure 11, that the quantities f_L , f_H and f_{CENTER} are the same as for those used to describe the behaviour of the previous band

pass filter. This is because the band stop filter is simply an inverted or complimented form of the standard band pass filter (BPF).

Again, the upper and lower cut-off frequency points for a band stop filter can be found using the same formula as that for both the low and high pass filters as shown.

$$f_c = \frac{1}{2\pi RC} \text{ Hz}$$

The geometric center frequency, f_{CENTER} as: $f_{\text{center}} = \sqrt{f_L \times f_H}$

The frequency at which this passive twin-T RC band stop filter design offers maximum attenuation within the notch is called the “notch frequency”, f_N and is defined as:

$$f_N = \frac{1}{4\pi RC} \text{ Hz}$$

Being a passive RC network, one of the main disadvantages of this basic twin-T band stop filter design is that the maximum value of the output (V_{OUT}) below the notch frequency is generally less than the maximum value of output above the notch frequency. This is due in part to the two series resistances ($2R$) of the low pass filter section having greater losses than the reactances of the two series capacitors (C) of the high pass section.

As well as uneven gains either side of the notch frequency, another disadvantage of this basic design is that it has a fixed Q value of **0.25**, in the order of **-12dB**. This is because at the notch frequency f_N , the reactances of the two series capacitors equals the resistances of the two series resistors, resulting in the currents flowing through each RC branch being out-of-phase by 180° .

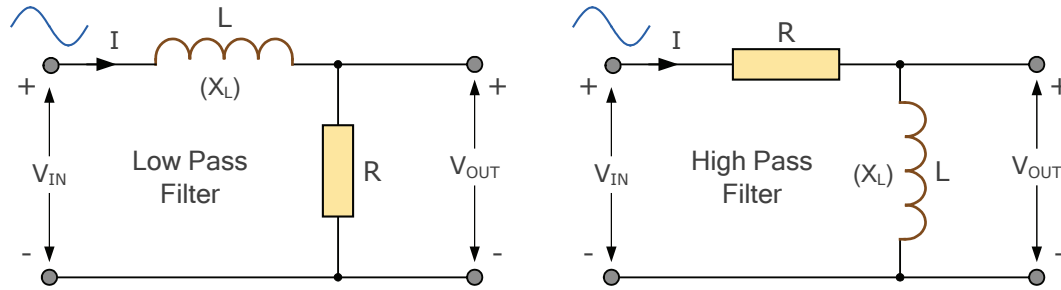
6. PASSIVE RL FILTERS

Inductors, coils and windings are widely used in electronic circuits, for filtering and/or phase shift applications. But because inductors are larger, heavier, and more expensive than capacitors, they are often less used for electronic filter designs.

While it maybe convenient to use capacitors to construct the various filter topologies seen previously, sometimes it is necessary to take advantage of the inductive reactance properties of coils and windings to construct resistor-inductor, **RL** filters or resistor-inductor-capacitor, **RLC** filters as coils provide a direct path for DC currents to flow.

Series **RL** (resistor-inductor) networks can be used as low pass and high pass filters as shown in Figure 12.

FIGURE 12. PASSIVE RL FILTER CIRCUITS



For the *low pass filter* circuit on the left, the input signal is applied across the series inductor and resistor network. The output signal is taken from across the resistor.

At low frequencies, the inductive reactance, X_L of the coil is very low allowing current, (I) to pass, therefore most of the voltage is dropped across the resistor. As the input frequency increases, the inductive reactance of the coil also increases proportionally offering more opposition to the flow of current. Thus, more voltage is dropped across the inductor and therefore proportionally less voltage is dropped across the resistor. Increasing the input frequency further decreases the output voltage, (V_{OUT}) even more with the series **RL** circuit acting as a low pass filter.

For the high pass filter circuit on the right, the input signal is applied across the resistor and inductor network, with the output signal is taken from across the inductor.

Passive RLC filters are commonly used in loudspeaker boxes

At low frequencies the inductive reactance, X_L is again low meaning that the inductor is effectively a short circuit so less voltage is dropped across the coil with more voltage dropped across the resistor. Thus the output is essentially zero at low frequencies.

At high frequencies, the inductive reactance of the coil becomes high, causing most of the voltage to be dropped across the coil. Increasing the input frequency further increases the output voltage, (V_{OUT}) even more with the series **RL** circuit acting as a high pass filter.

As with the **RC** filter circuits, the frequency above or below the frequencies passed or attenuated by the two **RL** filter networks is called the corner or cut-off frequency. The corner frequency for a series **RL** filter network is given as:

$$f_c = \frac{R}{2\pi L} \text{ (Hz)}$$

This corner frequency, f_c , can be set to any desired value by selecting the required values for **R** and **L** allowing us to design a passive low pass or high pass filter with whatever corner frequency is needed.

7. THE ACTIVE LOW PASS FILTER

As we have seen, a basic first-order passive filter circuit, such as a low pass or high pass filter, can be constructed using a single resistor in series with a capacitor connected across an input signal. The main disadvantage of passive filters is that the amplitude of their output signal will always be less than that of the input signal. That is their voltage gain is never greater than 1 (unity).

With passive filter circuits containing multiple stages, this loss in signal amplitude called “Attenuation” can become quite severe. One way of restoring or controlling this loss of signal is by using amplification through the use of *Active Filters*.

As their name implies, **Active Filters** contain active components such as operational amplifiers, transistors or FET's within their circuit design. They draw their power from an external DC power source and use it to boost or amplify their output signal.

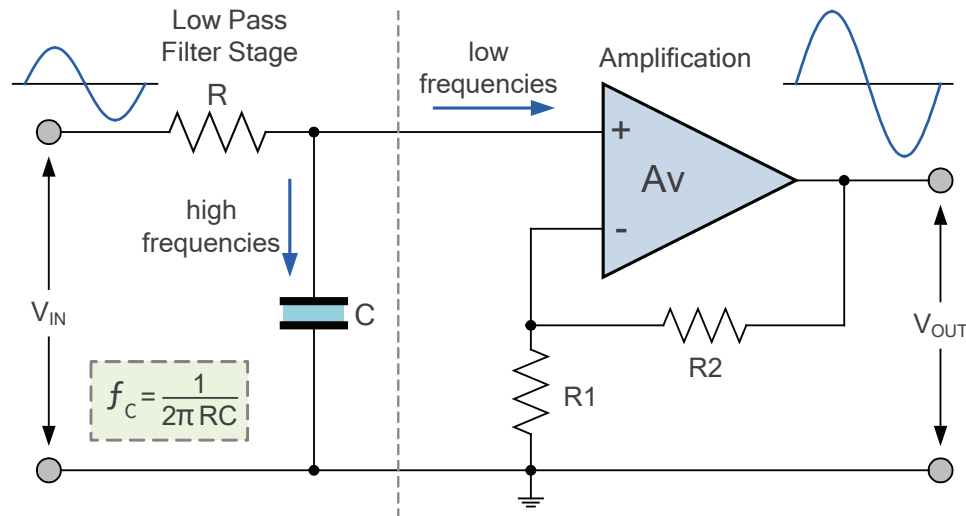
However, unlike a passive high pass filter which has in theory an infinite high frequency response, the maximum frequency response of an active filter is limited by the Gain/Bandwidth product (or open loop gain) of the operational amplifier being used.

Still, active filters are generally much easier to design than passive filters, they produce good performance characteristics over a range of frequencies, have very good accuracy with a steep roll-off and low noise when used with a good circuit design.

The most common and easily understood active filter design is again that of the **Active Low Pass Filter (LPF)**. Its principle of operation and frequency response is exactly the same as those for the previously seen passive low pass filter, the only difference this time is that it uses an operational amplifier to produce amplification and gain control.

The simplest form of a low pass active filter is to connect an inverting or non-inverting amplifier to the basic RC low pass filter circuit as shown in Figure 13.

FIGURE 13. ACTIVE LOW PASS RC FILTER CIRCUIT



The frequency response for the low pass circuit of Figure 13. will be the same as that for the passive RC filter. The amplitude of the output signal is increased by the pass band voltage gain, A_v of the amplifier. For the non-inverting operational amplifier circuit above, its DC voltage gain is given as a function of the feedback resistor (R_2) divided by its corresponding input resistor (R_1) value and is defined as:

$$A_v = 1 + \frac{R_2}{R_1}$$

The active low pass filter has a constant voltage gain A_v from DC (0Hz) up to its corner frequency point, f_c . At f_c , $R = X_C$ so the output voltage gain becomes equal to $0.707A_v$. After f_c the output decreases at a constant rate as the frequency increases.

When dealing with active filter circuits the magnitude of the pass band gain is generally

expressed in decibels, (dB) as a function of the voltage gain, and this is defined as being:

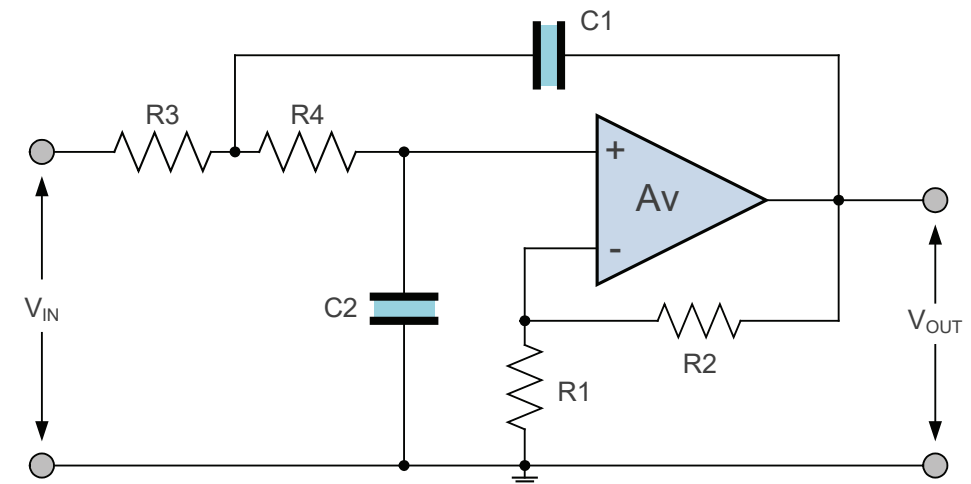
$$A_{V(dB)} = 20 \log_{10} \left(\frac{V_{OUT}}{V_{IN}} \right)$$

Thus at the corner frequency point f_c , the filters output gain will be **-3dB** ($20 \log(0.707)$).

8. SECOND ORDER ACTIVE LOW PASS FILTER

The advantage of using operational amplifiers within an electronic filter design, is that we can create higher order filters with very steep roll-off's into the stop band. First order active filters such as Figure 13. can be easily be converted into second order filters simply by adding an additional RC network within the input or feedback path. Then we can define second order active filters as simply being: "two 1st-order filters cascaded together with amplification" as shown in figure 14.

FIGURE 14. ACTIVE 2ND-ORDER LOW PASS FILTER CIRCUIT



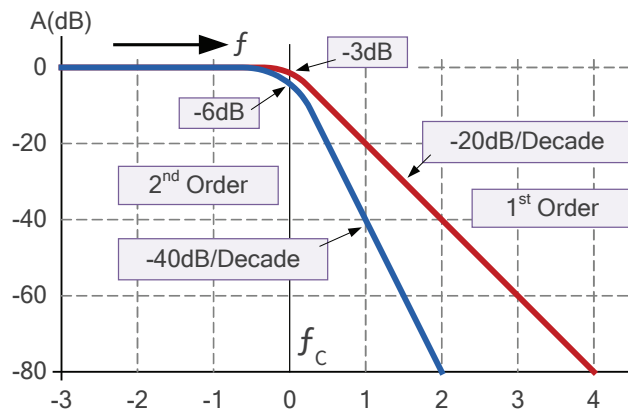
This second order low pass filter circuit of Figure 14 shows two RC networks, R3 – C1 and R4 – C2 which gives the filter its frequency response characteristics. The frequency response of the second-order low pass filter is identical to that of the first-order type except that the stop band roll-off will be twice the first-order filters at 40dB/decade.

Therefore, the design steps required of the second-order active low pass filter are the same. If the two resistors are of different values, and the two capacitors are of different values. The corner frequency point is defined as:

However, if resistors $R_3 = R_4$ and capacitors $C_1 = C_2$, we can use the standard equation of $f_c = 1/(2\pi RC)$ as before.

$$f_c = \frac{1}{2\pi \sqrt{R_3 R_4 C_1 C_2}}$$

FIGURE 15. 1ST AND 2ND-ORDER ROLL-OFF COMPARISON



The frequency response bode plot of Figure 15 shows the comparison between a 1st order and 2nd order roll-off. The difference is the steepness of the roll-off which is -40dB/decade in the stop band.

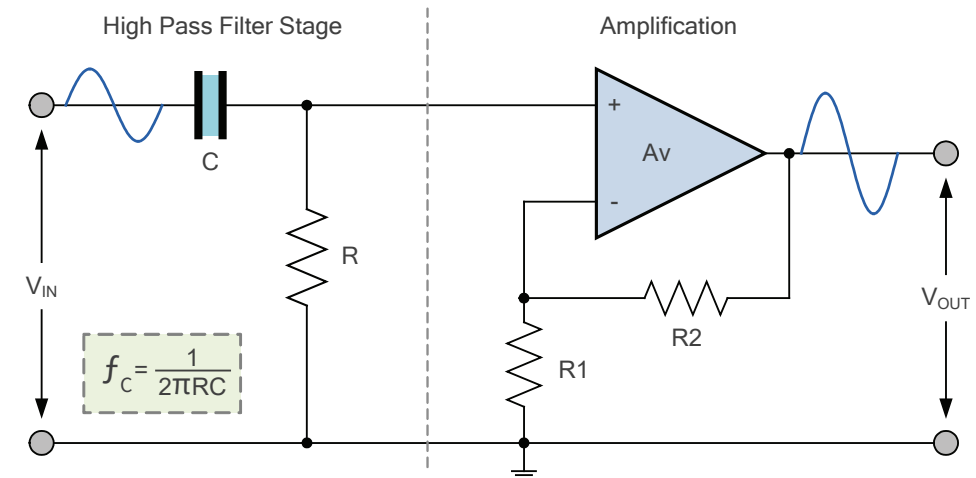
So, an active filter which has an n^{th} number order will therefore have a subsequent roll-off rate of $20n$ dB/decade. So a first-order filter is $20 \times 1 = 20$ dB/decade, a second-order filter is $20 \times 2 = 40$ dB/decade, and a third-order filter is $20 \times 3 = 60$ dB/decade, and a fourth-order filter is $20 \times 4 = 80$ dB/decade, etc.

9. THE ACTIVE HIGH PASS FILTER

The basic operation and frequency response of an **Active High Pass Filter (HPF)** is exactly the same as that for its equivalent RC passive high pass filter circuit of Figure 6. However, unlike the passive high pass filters which has an “infinite” frequency response, the maximum pass band frequency response of an active high pass filter is limited by the open-loop characteristics of the operational amplifier being used.

Just like the previous *active low pass filter circuit*, the simplest form of an active high pass filter is to connect a standard inverting or non-inverting operational amplifier to the basic RC high pass passive filter circuit as shown in Figure 16.

FIGURE 16. ACTIVE HIGH PASS RC FILTER CIRCUIT



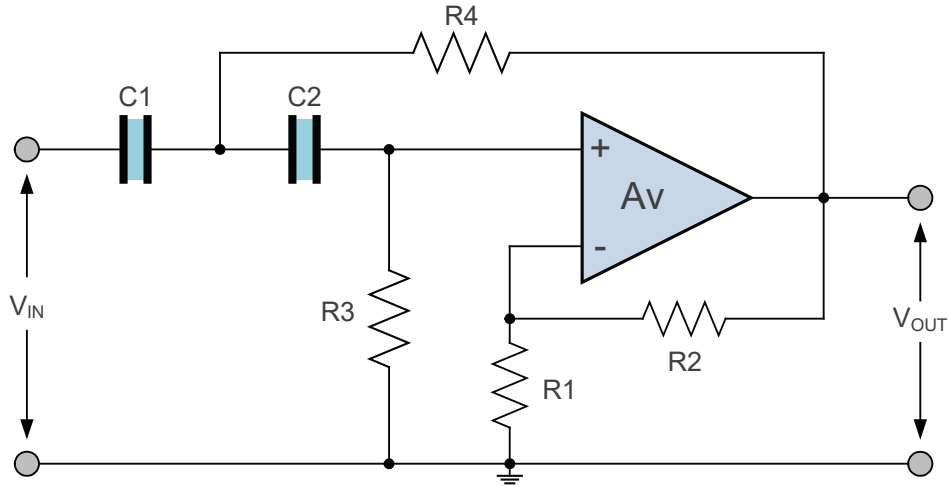
The frequency response of the active high pass filter is the same as that for the passive filter, except that the amplitude of the signal is increased by the gain of the amplifier. For a non-inverting amplifier the value of the pass band voltage gain is given as $A_v = 1 + (R_2/R_1)$. The same as for the low pass filter circuit.

The filter has an output voltage gain A_v which increases from DC (0Hz) to its corner frequency point, f_c at 20dB/decade as the frequency increases. At f_c when $X_c = R$, the output voltage gain becomes equal to $0.707A_v$ as before. Above f_c all input frequencies are pass band frequencies so the filter produces a constant gain.

10. SECOND ORDER ACTIVE HIGH PASS FILTER

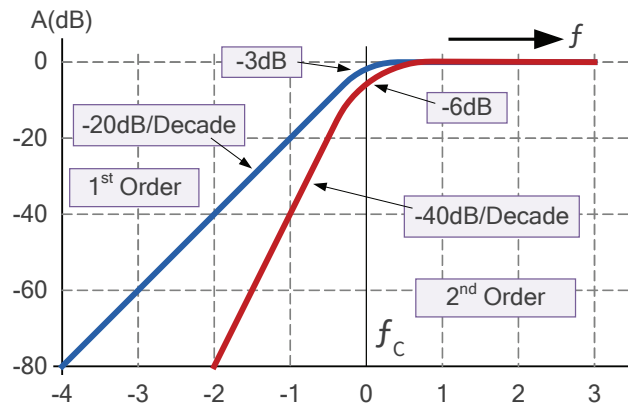
As before, an active high pass filter can be converted into a second-order high pass filter simply by using an additional RC network in the input and feedback path as shown in Figure 17. The frequency response of the second-order high pass filter is identical to that of the first-order type except that the stop band roll-off will be twice the first-order filters at 40dB/decade.

FIGURE 17. ACTIVE 2ND-ORDER HIGH PASS FILTER CIRCUIT



Since second-order high pass and low pass active filters are the same circuits except that the positions of the resistors and capacitors are interchanged, the design and frequency scaling procedures for the high pass filter are exactly the same as for those of the previous low pass filter circuit. Figure 18 shows the filters frequency response.

FIGURE 18. HIGH PASS 1ST AND 2ND-ORDER COMPARISON



The frequency response bode plot of Figure 18 shows the comparison between a 1st order and 2nd order roll-off at -40dB/decade from the stop band.

As before, the corner frequency point is calculated as:

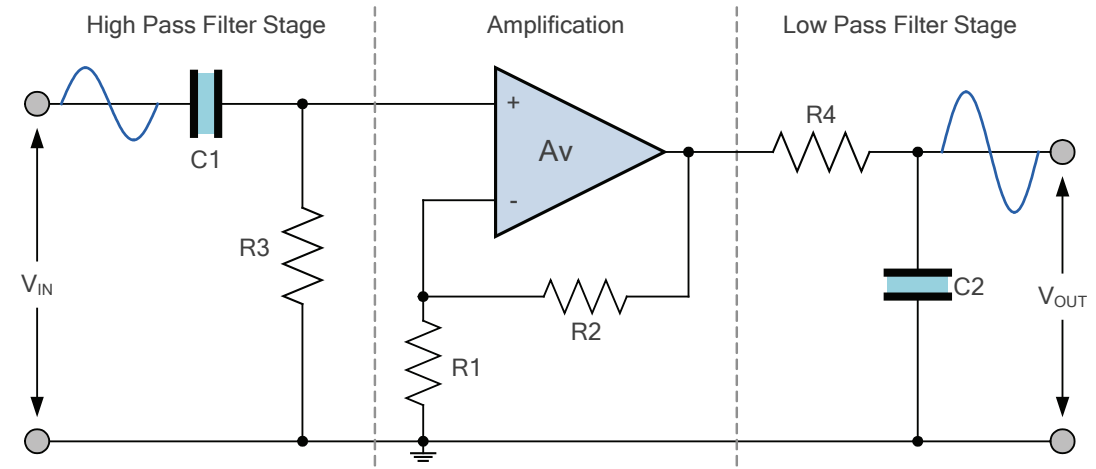
$$f_c = \frac{1}{2\pi \sqrt{R_3 R_4 C_1 C_2}}$$

If the resistors and capacitors have different component values.

11. ACTIVE BAND PASS FILTER

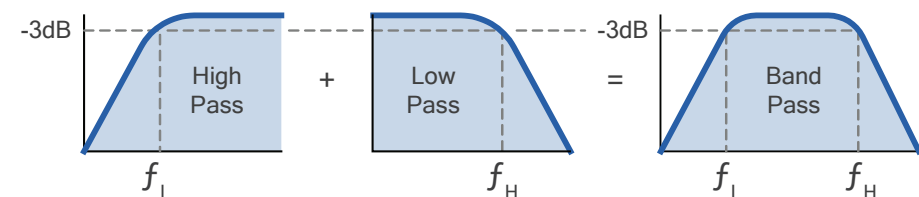
As seen previously, the *Band Pass Filter* passes a range of input signals within a certain “band” of frequencies between two corner frequency points. A simple **Active Band Pass Filter** can be easily made by cascading together a single *Low Pass Filter* with a single *High Pass Filter* as shown in Figure 19.

FIGURE 19. ACTIVE BAND PASS RC FILTER CIRCUIT



The difference between the -3dB corner frequency of the low pass filter and the -3dB corner frequency of the high pass filter determines the bandwidth (BW) of the pass band. The first stage of the filter will be the high pass stage that uses the capacitor to block any DC biasing from the source. This design has the advantage of producing a relatively flat asymmetrical but wide pass band frequency response as shown in Figure 20.

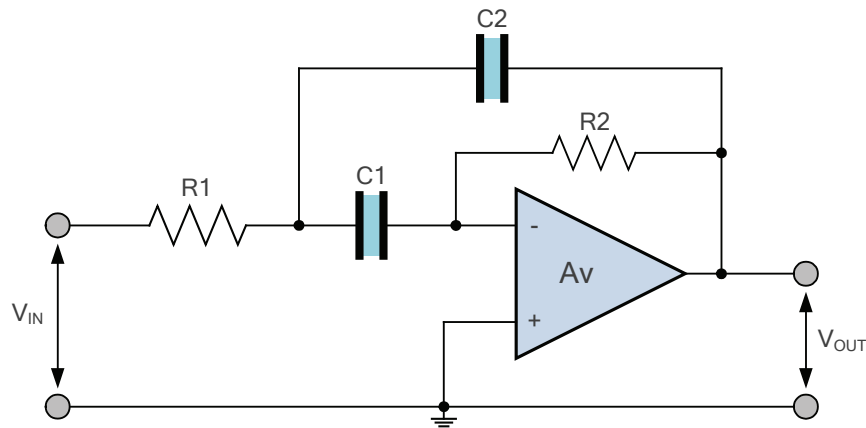
FIGURE 20. ACTIVE BAND PASS FILTER RESPONSE



The higher corner frequency point (f_H) as well as the lower corner frequency point (f_L) are calculated the same as before in the standard first-order low and high pass filter circuits. Obviously, a reasonable separation is required between the two corner frequencies to prevent any interaction between the two filter stages. The amplifier provides isolation between the two stages and defines the overall voltage gain of the pass band.

We can improve the pass band response of the filter circuit in Figure 19. by rearranging the RC components to produce an **infinite-gain multiple-feedback (IGMF)** band pass filter circuit. This type of active band pass design produces a “tuned” circuit based around a negative feedback active filter giving it a high Q-factor. Q-factor is a measure of how selective or non-selective the band pass filter is towards a given range of frequencies.

FIGURE 21. INFINITE GAIN MULTIPLE FEEDBACK ACTIVE BAND PASS FILTER



This active band pass filter circuit of Figure 21. uses the full gain of the op-amp with multiple negative feedback applied via resistor, **R2** and capacitor **C2**. This produces an amplitude response and steep roll-off on either side of its center frequency.

Because the frequency response of the circuit is similar to a resonance circuit, this center frequency is usually referred to as the circuits resonant frequency, (f_r).

Then we can define the frequency response characteristics of the IGMF filter as follows:

$$f_r = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad Q_{BP} = \frac{f_r}{BW_{(-3dB)}} = \frac{1}{2} \sqrt{\frac{R_2}{R_1}}$$

$$\text{Maximum Voltage Gain, } (A_V) = -\frac{R_2}{2R_1} = -2Q^2$$

We can see then that the relationship between resistors, **R1** and **R2** determines the band pass “Q-factor” and the frequency at which the maximum amplitude occurs, the gain of the circuit will be equal to **-2Q²**. Then as the gain increases so to does the selectivity. In other words, high gain = high selectivity.

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